

● EASY

● GEOMETRY AND TRIGONOMETRY

Area and volume

01

10 questions · ~15 min

Q1

GEOMETRY AND TRIGONOMETRY

What is the perimeter, in inches, of a rectangle with a length of 4 inches and a width of 9 inches?

- A. 13
- B. 17
- C. 22
- D. 26

Triangle ABC and triangle DEF are similar triangles, where $\overline{AB}$ and $\overline{DE}$ are corresponding sides. If $\overline{DE} = 2\overline{AB}$ and the perimeter of triangle ABC is 20, what is the perimeter of triangle DEF ?

- A. 10
- B. 40
- C. 80
- D. 120

The perimeter of triangle ABC is 17 inches, the length of side AB is 4 inches, and the length of side AC is 7 inches. What is the length, in inches, of side BC ?

- A. 4
- B. 6
- C. 7
- D. 11

A rectangle has a length of 3 units and a width of 39 units. Which expression gives the area, in square units, of this rectangle?

A. $23 + 39$

B. $23 \cdot 39$

C. $3 + 39$

D. $3 \cdot 39$

Q5

GRID-IN

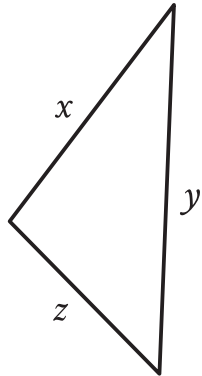
GEOMETRY AND TRIGONOMETRY

A rectangle has a length of 64 inches and a width of 32 inches. What is the area, in square inches, of the rectangle?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



Note: Figure not drawn to scale.

- The lengths of the sides are x , y , and z .
- A note indicates the figure is not drawn to scale.

The triangle shown has a perimeter of 22 units. If $x = 9$ units and $y = 7$ units, what is the value of z , in units?

- A. 6
- B. 7
- C. 9
- D. 16

Q7

GRID-IN

GEOMETRY AND TRIGONOMETRY

A right rectangular prism has a length of 11 meters, a width of 8 meters, and a height of 10 meters. What is the volume, in cubic meters, of the prism?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

What is the area, in square centimeters, of a rectangle with a length of 36 centimeters and a width of 34 centimeters?

- A. 70
- B. 140
- C. 1,156
- D. 1,224

What is the area of a rectangle with a length of 4 centimeters cm and a width of 2 cm?

- A. 6 cm²
- B. 8 cm²
- C. 12 cm²
- D. 36 cm²

The area of a rectangle is 57 square inches. The length of the longest side of the rectangle is 19 inches. What is the length, in inches, of the shortest side of this rectangle?

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.